Project

VLSI Class 2026

Design Example

Follow the instructions here to set up a virtual machine (VM) with the SKY130 PDK and the required EDA tools pre-installed.

Next, study the basic concepts through this practical design example. You are not required to re-implement the example; simply clone the example project into the VM and explore it. Make sure you understand the design flow, the underlying concepts, and how to use the associated tools.

Topic

Design a 16-bit Zero Leading Counter (ZLC) circuit using the complete design flow demonstrated in the design example.

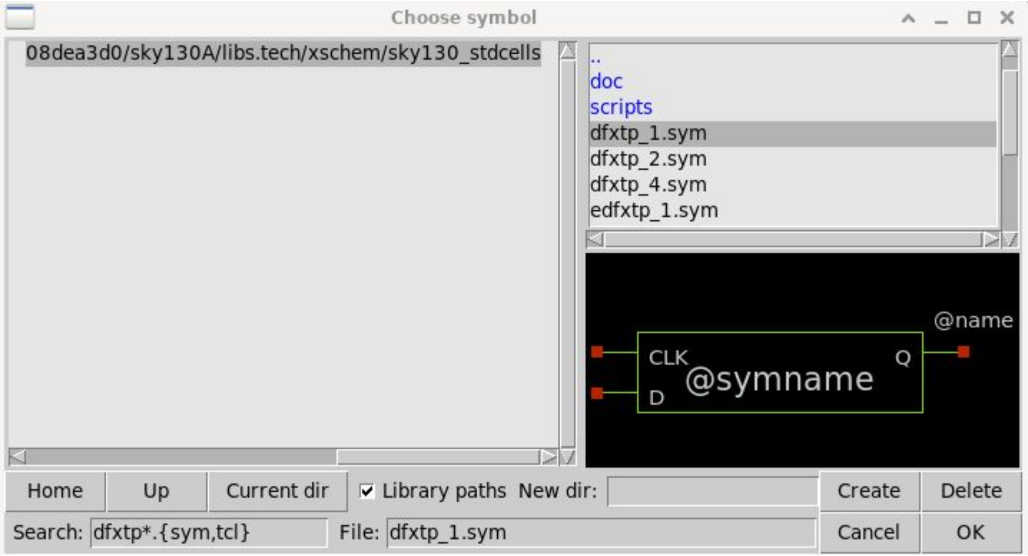
A Zero Leading Counter counts the number of consecutive zeros starting from the most significant bit (MSB) of the input word. For example, the 16-bit input 16'b0001011000100000 contains three leading zeros, so the output should be 5'b00011.

If you implement the design as a combinational circuit, the module should include: a 16-bit input A and a 5-bit output Y.

If you implement the design as a sequential circuit, you may introduce additional signals as needed, such as clk, data_ready, or similar control signals.

Requirement (1/2)

You may use any gates available in the SKY130 standard cell library in your design. If a flip-flop is required, use **dfxtp_x**, which is a positive-edge-triggered D flip-flop without a reset input.



Requirement (2/2)

Do not cheat. All design steps must be performed manually by your team, following the same methodology presented in the design example.

Your circuit must successfully pass functional simulation, DRC, and LVS verification.

Your circuit will be evaluated using the same timing and analysis settings in the adder_4bit.tcl script.

The final layout should have a reasonably square aspect ratio, with all input and output ports routed to the boundaries of the layout.

Designs with better optimization in terms of area, timing performance, and power consumption will receive higher scores.

Report (1/2)

The report must be submitted in PDF format and include the following contents:

A description of the circuit functionality, interface, algorithm, and overall architecture.

A complete presentation of the design flow, following the same structure as the design example.

Explanations and justifications for your design choices and implementation decisions.

A discussion of any optimization strategies used to improve area, timing performance, or power consumption.

A final results section (last page in the report) summarizing the completed design and evaluation metrics (details provided on the next slide).

Report (2/2)

The final result section must include a table that summarizes your circuit metrics:

Throughput (Mops/s)	Max clock rate (MHz)	Power @ 1 Mhz (μW)	Power @ max clock (μW)	Layout area (μm²)	Layout aspect ratio (W:H)
50	100	35	140	2000	1.5:1

<u>Note</u>:

Throughput refers to the maximum number of operations the circuit can perform per second. For example, if a circuit can operate at a maximum clock frequency of 100 MHz, and it processes one input per clock cycle, then its throughput is 100 Mops/s.

The maximum clock frequency should be estimated using STA and verified through post-layout simulation, with appropriate explanation and justification of the results.

The layout aspect ratio is the ratio between the layout width and height.

Files to submit

The following files must be organized in a directory structure similar to that used in the design example:

Xschem schematics (.sch) and symbols (.sym)

Magic layout files (.mag)

Ngspice testbenches (.spice)

Icarus Verilog testbenches (.v)

Compress the top-level project directory into a **.zip** file and submit it together with the report.